

## # Motivation:-

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Due to two principal application areas:

- 1) Improvement of pictorial information for human interpretation
- 2) Processing of image data for storage, transmission and representation for autonomous machine perception.

## # Introduction:-

Image - An image is a projection of a 3D scene into a 2D projection plane.



- An image is defined as a two-dimensional function  $f(x, y)$  where  $x$  and  $y$  are spatial coordinates, and the amplitude of  $f$  at any pair of coordinates  $(x, y)$  is called the intensity or gray level of the image at that point.
- So depending upon  $x$ ,  $y$ , and  $f$  values, image can be classified into two types:

1) Analog Image: It can be mathematically represented as a continuous range of values representing position  $(x, y)$  and intensity  $(f)$ .

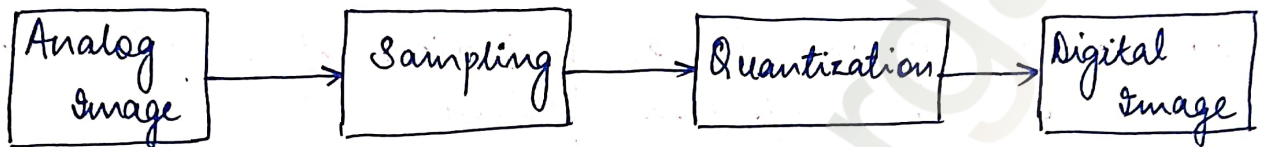
eg Image produced on the screen of a CRT monitor.

Note: Storage requirement requires high memory.

2) Digital Image - When  $x, y$  and  $f$  are all finite, discrete quantity, image is a digital image.

Note:- Generally, everything is in the analog domain, but for storage or transmission the image must be converted to digital domain with help of digital computer.

- Conversion of an analog image into digital image involves two main operations, sampling & quantization.



Sampling: discretization of  $x, y$

Quantization: discretization of  $f$  parameter

- Digital image contains finite number of elements, each of which has particular location and value. These elements are called picture elements, image elements, pels or pixels.
- Advantages
  - Fast processing
  - Cost Effective
  - Effective Storage
  - Efficient Transmission
  - Scope for versatile image transmission
- Disadvantages
  - High memory required to store & process good quality images.
  - Processor has to be very fast because volume of data is very high.

## Digital Image Processing : (DIP)

(2)

- The analysis and manipulation of a digitized image, especially in order to improve its quality is known as DIP.
- The processing of digital images by means of digital computers are known as DIP. <sup>(OR)</sup>
- Advantages
  - Humans are limited to the visual band of electromagnetic (EM) spectrum.
  - But imaging machines covers almost the entire EM spectrum ranging from gamma to radio waves.
  - Therefore, operation on the images generated by the sources that humans are not capable to sense. This can be done by DIP.
  - This include ultrasound, electron microscopy, computer generated images. Therefore, DIP includes wide variety of applications.
- Scope
  - There are fields like computer vision which use computers to emulate human vision, including learning and being able to make inferences and take actions based on visual input.
  - Therefore, is a branch of Artificial Intelligence (AI) whose objective is to emulate human intelligence.
  - Hence, the area of image analysis, is in between image processing and computer vision.



## Useful Paradigm -

For continuum from image processing to computer vision are

→ Low-level Processes -

Image pre-processing to reduce noise, contrast enhancement and image sharpening.

→ Mid-level Processes -

Segmentation, Classification

→ High-level Processes -

Image Analysis.

## Origin of DIP:-

- The first application of digital images was in the newspaper industry.
- In this application, the pictures were sent by submarine cable between two cities.
- Invention of the Bartlane cable picture transmission system in the early 1920's reduced the time required to send a picture across the Atlantic from more than a week to less than a three hours.
- Specialized printing equipment coded pictures for cable transmission and then reconstructed them at the receiving end.
- The early Bartlane systems were capable of coding images in five distinct levels of gray. This capability was increased to 15 levels in 1929.

Note:- Fact is history of DIP is tied to the development of the digital computer.

- ③
- The first powerful computer to carry IP appears in 1960's.
  - Parallelly, for space applications, in late 1960 and early 1970, DIP is used in medical imaging, remote sensing, and astronomy.
  - The invention of Computerized Tomography (CT) in 1970's is the important application of IP in medical diagnosis.
  - From 1960's to present, the field of IP grows rigorously in the field of industry, medicines, biological sciences, law enforcement, defence etc.

### Applications

Digital Image Processing & Analysis techniques are used today in variety of applications such as -

#### ① Office Automation

- Optical Character Recognition
- Document Processing
- Identification of address on envelope
- Logo & Icon Identification

#### ② Industrial Automation

- Automatic inspection system
- Non destructive testing
- Process control applications
- Automatic assembling

#### ③ Bio - Medical

- ECG Analysis
- X-Ray Image Analysis
- Detection of Various Disease
- Mammograms, Tomography.

#### ④ Remote Sensing

- Estimation related to Agriculture
- Environment & Pollution Control.
- Monitoring traffic along roads & airfields
- Urban Planning

#### ⑤ Scientific Applications

- High Energy Physics
- Bubble Chamber
- Track Analysis

#### ⑥ Criminology

- Finger Print Identification
- Human Face Registration & Matching
- Forensic Investigation

#### ⑦ Astronomy & Space Applications

- Restoration of images suffering from geometric distortion
- Computing Pictures of Planetary Surfaces

#### ⑧ Meteorology

- Short Term Weather forecasting
- Climate Change detection from satellite
- Cloud Pattern Analysis

#### ⑨ Information Technology

- Facsimile Image Transmission
- Video Conferencing
- Video phones

#### ⑩ Entertainment & Consumer Electronics

- HDTV
- Multimedia
- Video Editing



## ⑪ Printing & Graphic Arts

- Colour fidelity in desktop publishing
- Art conservation & Dissemination

## ⑫ Military Applications

- Missile guidance & detection
- Target Identification
- Navigation of Vehicle
- Range finding

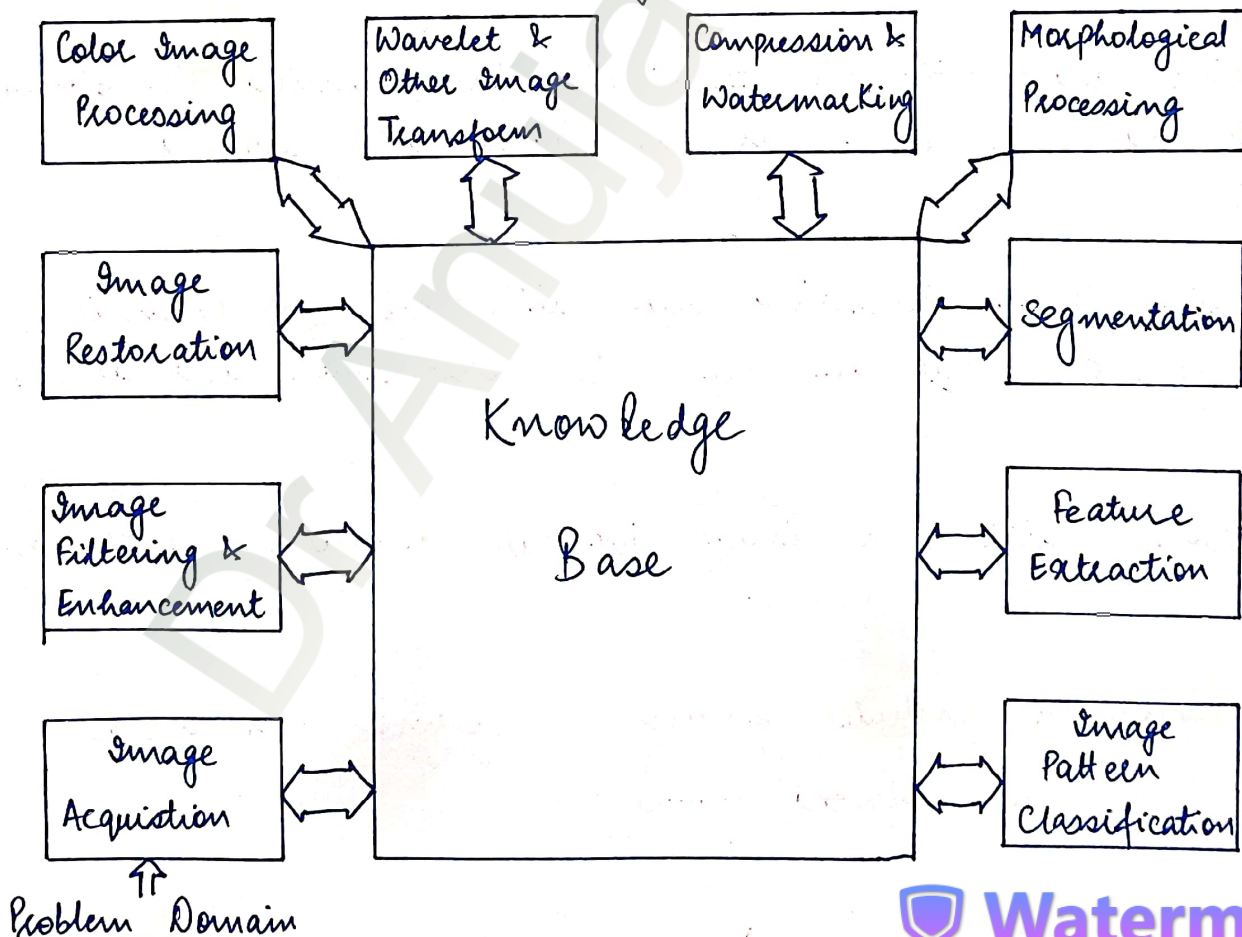
## ⑬ Security Measure

- Recognition of human objects & verification of identity based on images of face, iris, palm & fingerprint.

## ⑭ Image Database

- Organizing, indexing and efficient search of image document based on the content, digital library.

## # Fundamental Steps in Digital Image Processing



The basic fundamental steps of digital image processing are as follows -

- ① Acquisition could be simple as being given an image that is already in digital form. Generally, it involves preprocessing such as scaling.
- ② Image enhancement is the process of manipulating an image so the result is more suitable than the original for a specific application.
- ③ Image restoration deals with improving the appearance of an image. It is based on mathematical or probabilistic model of image degradation.
- ④ Color image processing is used for the basis of extracting features of interest in an image.
- ⑤ Wavelets are the foundation for representing images in various degree of resolution.
- ⑥ Compression deals with the techniques for reducing the storage required to save an image or the bandwidth required to submit it.
- ⑦ Morphological Image Processing deals with tools for extracting image components that are useful in representation and description of shape.
- ⑧ Segmentation partition an image into its constituent parts or objects.
- ⑨ Feature Extraction includes finding the features and assigns attributes to the detected features.



(10) Image Pattern Classification is the process that assigns a label to object based on its feature descriptors. The methods used are minimum distance, Bayes Classifier, Deep Neural Networks. (5)

Note: → Knowledge about a problem domain is coded into an image processing system in the form of knowledge database.

→ The knowledge base can be quite complex, such as interrelated possible defects or high resolution satellite images.

### # Components of Image Processing System

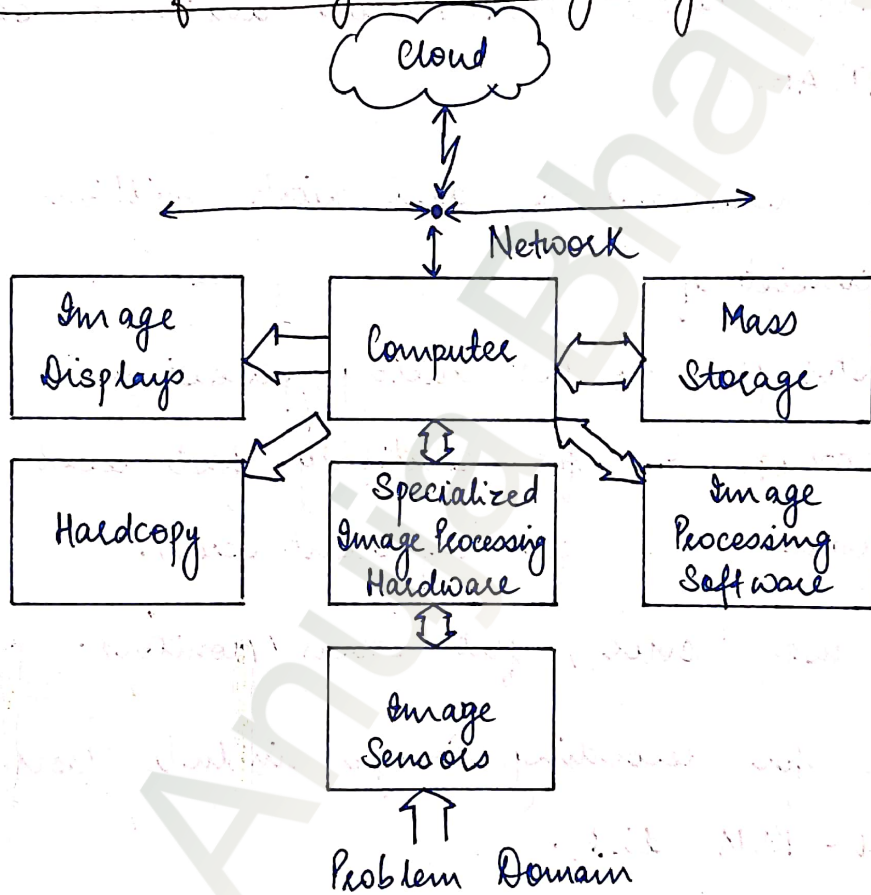


Figure shows a basic components consist of typical general purpose system used for digital image processing.

→ Two subsystems are required to acquire digital images.

1. Physical Sensor - responds to the energy radiated by the object we wish to image.

2. Digitizer - device for converting the output of the physical sensing device into digital form.

- Specialized image processing hardware usually consist of a digitizer plus hardware that performs ALU operations on entire image.
- Computer in an image processing system is a general-purpose computer and can range from a PC to supercomputer.
- Software for image processing consist of specialized modules that perform specific task. Commercially available software known as MATLAB.
- Mass Storage is necessary to deal with millions of image. Digital Storage consist of
  1. Short-Term Storage - use during processing.
  2. On-line Storage - for relatively fast recall.
  3. Archive Storage - for infrequent access.
- Image display use color, flat-screen monitors.
- HardCopy devices for recording images include laser-printer, film camera, CD-ROM disk.
- Networking and Cloud Communication are default in any computer system. Because of large amount of data in image processing applications, the key is to consider in image transmission is Bandwidth.

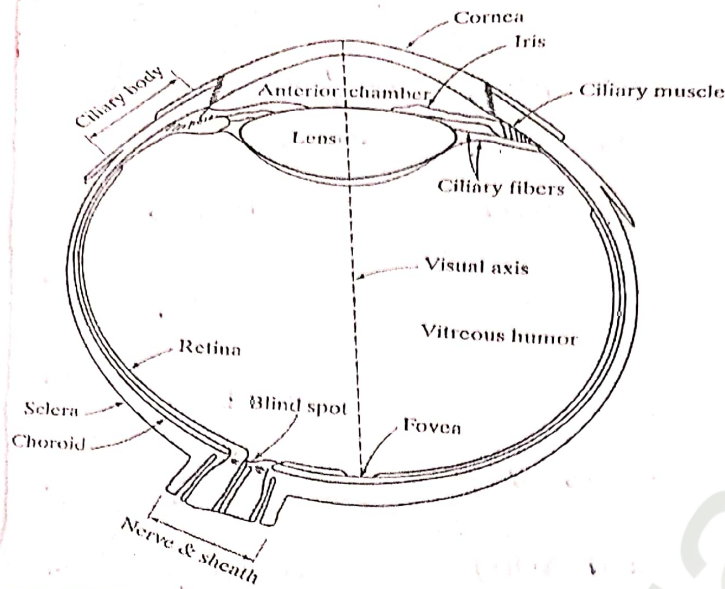


# # Human Visual System

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- It is important for designers and users of image processing to understand the characteristics of the human visual system.
- For efficient design of algorithms whose output is a image or display viewed by human observer, it is beneficial to have an understanding of human vision.
- Fig shows the horizontal cross-section of human eye-ball
  - The front of the eye is covered by transparent surface called cornea.
  - The remaining outer cover, sclera consist of choroid & blood capillaries.
  - Inside the choroid, is the retina which is composed of two type of photoreceptors
    - (i) Rods - long & slender
    - (ii) Cones - short & thicker
  - light entering the cornea is focused on retina surface by a lens that changes shape to perform proper focus of near & distant objects.
  - The iris control the amount of light entering the eye.
  - Rods are important for black & white vision in dim light. They allow us to discriminate between different shades of dark & light. It provide a visual response called Scotopic Vision. (see in the dark)





→ Cones provide colour vision and high vision in bright light. It responds to higher level of illumination called as photopic vision.

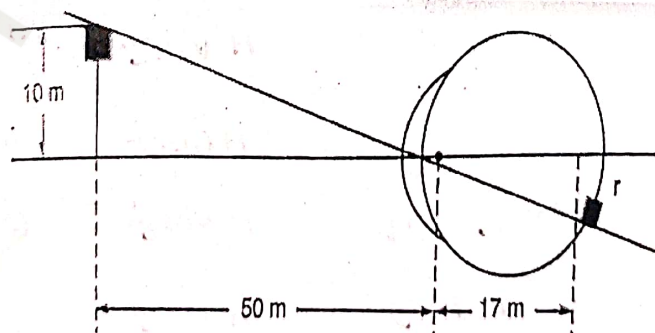
### # Image formation in an eye

- The distance between the centre of the lens and retina varies from 14 mm to 17 mm.

→ When the eye focus on object far away the lens exhibits (17 mm)

→ When the eye focus on near by objects, the lens (14 mm)

eg Consider an observer looking at lamp post at distance 50 m. If height of lamp is 10 m. Find size of image formed in retina.



Since the image is far away, the focal length is 17 mm. ⑦

∴ Similarity of triangles, let the retinal image be  $x$

$$\frac{10}{50} = \frac{x}{17} \Rightarrow x = 3.4 \text{ mm}$$

∴ Height of retinal image = 3.4 mm

→ If image is at distance 100 m, then

$$\frac{10}{100} = \frac{x}{17} \Rightarrow x = 1.7 \text{ mm}$$

## # Visual Phenomena

- The human eye sense w.r.t Intensity, Brightness and Contrast.

→ Intensity of a light source depends on the total amount of light emitted by it. Hence, intensity is a physical property and can be measured.

→ Brightness is a psycho-visual concept and hence is actually a sensation to light intensity.

→ Contrast may be defined as difference in perceived brightness

## # Image Sensing & Acquisition

(from PPT) ①

- # A Simple Image formation Model

- # Sampling & Quantization

} PPT ①

## # Spatial & Intensity Resolution

- Spatial resolution is a measure of the smallest discernible detail in an image.

It is commonly measure in line pair per unit distance or dots per unit distance. eg Newspaper 75 dpi

- Image resolution is the largest number of discernible line pair per unit distance.

- The spatial resolution must be stated with spatial units.

eg Image with resolution of  $1024 \times 1024$  pixel is not meaningful without mentioning the spatial dimension.

eg Digital Camera with 20 megapixel chip has high capability than 8 megapixel camera.

- Intensity Resolution refers to smallest discernible change in intensity level.

eg Number of intensity level generally measures as power of two

Most commonly used number is 8-bits, with 16-bits being used in some applications in which enhancement of specific intensity range is necessary.

eg 256 levels has 8-bit of intensity resolution.

Note:- The discernible change in intensity are influenced by noise and saturation values.